

Anh Thuan Tran

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SUMMARY

Ph.D. Student in Computer Science with research experience on Computer Vision, Deep Learning, 3D Gaussian Splatting, 3D vision, 3D point cloud understanding, and vision-language models.

EDUCATION

George Mason University

Ph.D in Computer Science

Virginia, USA

Sep 2024 – Exp. May 2029

- **Research Interest:** 3D Vision, 3D Gaussian Splatting, 3DGS SLAM, Dynamic 3D Reconstruction, Vision Language
- **Advisor:** Prof. Jana Kosecka

Pukyong National University

M.Eng. in Artificial Intelligence Convergence | GPA: 3.89

Busan, South Korea

Mar 2022 – Feb 2024

Vietnam National University

B.Ec. in E-commerce | GPA: 3.42 (Thesis track)

Ho Chi Minh City, Vietnam

Sep 2017 – Jul 2021

PUBLICATIONS

- [1] **Anh Thuan Tran** and Jana Kosecka. VarSplat: Uncertainty-aware 3D Gaussian Splatting for Robust RGB-D SLAM. In *CVPR*, 2026. [\[PDF\]](#) | [Project Page](#).
- [2] **Anh Thuan Tran** and Jana Kosecka. PointSplat: Efficient Geometry-Driven Pruning and Transformer Refinement for 3D Gaussian Splatting. In *Under Review (CVPRW)*, 2026. [\[PDF\]](#).
- [3] **Anh-Thuan Tran**, Hoanh-Su Le, Suk-Hwan Lee, and Ki-Ryong Kwon. PointCT: Point Central Transformer Network for Weakly-Supervised Point Cloud Semantic Segmentation. In *WACV*, pages 3556–3565, 2024. [\[PDF\]](#) | [Code](#).

RESEARCH EXPERIENCE

Artificial Intelligence and Database Lab

Graduate Research Assistant

GMU, USA

Sep 2024 – Present

- **VarSplat** (SLAM & Uncertainty): Built an uncertainty-aware RGB-D 3DGS-SLAM system, rendering per-pixel uncertainty via the law of total variance to improve long-sequence robustness.
- **PointSplat** (Pruning): Developed image-agnostic pruning and refinement for 3D Gaussians in medium-scale indoor scenes with Point Transformer network.
- **Dynamic 3DGS** (Ongoing): Developing continual learning method to update 3D Gaussian maps in changing environments while excluding distractors such as moving people.
- **Open-Vocabulary 3DGS**: Implemented vision-language baseline by integrating LSeg features or CLIP with DINOv2 and SAM into 3D Gaussian attributes.

- **PointCT**: Proposed a point-based Transformer network for weakly-supervised point cloud segmentation.
- Developed central-based attention block to integrate global features across neighborhoods, and improve unlabeled point representations.
- Achieved competitive performance on the S3DIS, ScanNet with 0.1% labeled points, STPLS3D with 0.01% labeled points.

WORK EXPERIENCE

Newgen I&S

AI Researcher

Busan, South Korea

Jan 2024 – Jul 2024

- Developed offline chatbots for marine industry using LangChain and Ollama to process bilingual (English–Korean) PDFs.
- Processed text with FAISS, updated PostgreSQL, and LLM responses with top-k retrieval.
- Deployed system on server with WebSocket for real-time communication.

SCHOLARSHIPS

- Graduate Teaching Assistant, George Mason University (Sep 2024 - Present)
- Brain Korea 21, Graduate Student Research Scholarship (Artificial Intelligence Convergence Research Group), Pukyong National University (Mar 2022 - Feb 2024)
- Full-time Scholarship For Graduate Student, Pukyong National University (Mar 2022 - Feb 2024)
- Graduate student scholarship, Pukyong National University (Sep 2022 - Feb 2023)

SKILLS

- **Programming:** Python — Proficiency: Advanced
- **ML & 3D Vision:** 3D Gaussian Splatting, NeRFs, 3D Vision, Point Clouds, 3D Point Transformers, 2D–3D Distillation, Vision–Language Models, Open-Vocabulary Understanding
- **Frameworks & Libraries:** PyTorch, PointOps, NumPy, OpenCV, CLIP, DINO, NeRFStudio, Differentiable Gaussian Rasterization with Depth, Pose, and Variance.
- **Tools & Systems:** Ubuntu, Slurm, Blender, LangChain, Ollama, PostgreSQL, FAISS, RAG.

AWARDS

- Outstanding International Student Award - Pukyong National University (2024)
- Excellent achievement in undergraduate research – VNU-HCM (2022)
- Third Prize Young Scientist – University of Economics and Law, VNU-HCM (2018, 2020)